

CII & EEXI Compliance Strategies

Practical Assessment Methodology for Bulker Energy Efficiency Retrofits and CII Trajectory Optimization

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1. Executive Summary

With the enforcement of the International Maritime Organization (IMO) Carbon Intensity Indicator (CII) and Energy Efficiency Existing Ship Index (EEXI), commercial vessel operators face mounting operational penalties and chartering discounts for vessels rated in Grade D or E. This paper presents an integrated engineering framework for commercial bulk carriers to maintain Sustainable Grade B/A performance over a 5-year operational horizon through cost-effective energy-saving retrofits.

5% – 12%

HYDRODYNAMIC ESD SAVINGS

12 - 22 Mos

AVERAGE CAPEX PAYBACK

Grade B

TARGET CII TRAJECTORY

2. CII Degradation Mechanisms & Baseline Modeling

Vessel Operational Carbon Intensity (AER/cOI) naturally degrades by 1.5% to 3.0% annually due to hull fouling, propeller surface degradation, main engine mechanical wear, and increasing auxiliary electrical loads. Without intervention, a standard Supramax/Handymax bulker operating on VLSFO will transition from Grade C to Grade E within 24–36 months under tightening IMO annual reduction factors.

Retrofit Technology	Primary Mechanism	Est. Fuel Reduction	CII Grade Impact
Propeller Boss Cap Fins (PBCF)	Hub vortex cavitation cancellation	2.0% – 3.5%	+0.5 Grade step
Pre-swirl Duct (Mewis Type)	Wake equalizing & thrust recovery	4.5% – 7.5%	+1.0 Grade step
Silicone Foul-Release Coating	Hydrodynamic drag minimization	4.0% – 6.0%	+1.0 Grade step
Variable Frequency Drives (VFD)	Dynamic cooling pump control	30% – 50% (Aux Load)	+0.5 Grade step

3. Recommended Engineering Roadmap

To ensure long-term compliance without sacrificing operational speed or charter marketability, Fleet Experts Solutions recommends a phased upgrade schedule aligned with scheduled dry-dockings:

- Phase 1: High-ROI Hydrodynamic Upgrades** — Retrofit Mewis Ducts and PBCF during scheduled dry-docking. Re-align propulsion engine load diagram.

2. **Phase 2: Auxiliary Load Optimization** — Install frequency converters on Main Engine Seawater Cooling Pumps and Engine Room Supply Fans.
3. **Phase 3: High-Performance Hull Hydrodynamics** — Apply fluoropolymer/silicone foul-release coatings combined with ultrasonic anti-fouling transducers in sea chests.

Key Conclusion: Combining ESD installation with targeted operational speed optimization guarantees sustained Grade B performance through 2030 while significantly lowering daily fuel expenditures.

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